

MLB Batting Performance Analysis Report

Project Focus

This analysis focuses on predicting batter performance using MLB batting statistics from 2015 to 2024.

Methodology Summary and Results

EDA and Preprocessing

The dataset included many players who don't bat or players who don't bat often. We excluded these players to minimize the influence of luck on our stats, which could introduce noise.

We created a 4-tier classification system using **wRC+** thresholds.

After preprocessing, we were left with 2865 rows and 285 columns.

Outlier Detection

- Isolation Forest: Effective with highly dimensional data, which we have.
- LOF: Detect local anomalies by comparing local densities of neighbourhood.

Using a 5% contamination level, both methods identified 144 outliers, with 48 overlapping.

Top outliers included MVP-caliber players like Aaron Judge (220 wRC+), bottom outliers included struggling hitters (~30 to 40 wRC+), and there were some in the middle, such as Joey Gallo, who is notorious for being an anomaly in the baseball world.

We decided to **keep** outliers, as they represent real performances, rather than erroneous data.

Feature Selection

Before applying feature selection methods, we removed some features using domain knowledge: eliminated counting stats and kept rate-based stats; removed value-based stats, which leak target variable info; highly correlated stats; etc.

- RFE: Selected the 5 most important features (AVG+, OBP+, SLG+, HR/FB%, xSLG).
- Lasso Regression: Selected 59 important features.
- Manual Selection: Selected 13 features commonly used by domain experts.

Clustering

- K-Means: Implemented as baseline to observe performance with different k 's. Evaluated with silhouette scores and elbow method.
- Hierarchical: Dendrogram visualization to see hierarchical structure with different k 's.
- HDBSCAN: Automatically find clusters of different shapes and densities, as well as number of clusters.

RFE features outperformed other selection methods with a ~0.35 silhouette score using each clustering method, and captured more variance in 2D PCA.

Clustering performed poorly on our data due to its continuous nature, as we got weak silhouette scores using each method. Inspecting stats within clusters provided more insights, allowing us to spot potential player archetypes (discussed later).

Classification

We applied SMOTE to address class imbalance before applying classification methods, as Elite players made up a small portion of the data.

- Random Forest: Handles high-dimensionality, robust to multicollinearity, captures non-linear relationships, easy to interpret, etc.
- XGBoost: Common in baseball statistics, good with tabular data, computationally efficient, etc.

XGBoost outperformed Random Forest without feature selection (82.4% vs 89.9% accuracy).

Random Forest and XGBoost with RFE were comparable (88.0% vs 88.1% accuracy).

Hyperparameter Tuning

Tuning greatly improved the performance of Random Forest without feature selection (from 82.4% to 89.9%); however, all other models worked great out-of-the-box, before tuning.

After tuning, the best models were Random Forest and XGBoost, both without feature selection, and both with an accuracy of 89.9%.

Models with RFE initially outperformed no feature selection, but after tuning, models without feature selection outperformed RFE. This suggests that although some features are not as important, they can still play a part in predicting performance.

Domain Specific Insights

The continuous nature of baseball data makes it difficult to cluster data (silhouette < 0.4). Players succeed through all sorts of combinations of skills, making natural groupings difficult to create. However, inspecting stats within clusters ($k = 4$) revealed key information. After plotting stats of each cluster using radar charts, we found various hitter archetypes: power hitters, contact hitters, all-around stars, etc.

RFE revealed the most important stats for predicting wRC+ include AVG+, OBP+, and SLG+, which are commonly referred to as the “Triple Slash,” and serve as a quick way to gauge a hitter’s ability. It also selected HR/FB%+ and xSLG, both reflecting a hitter’s power ability, which has defined how modern hitters approach the game.

Challenges Encountered

We used wRC+ as the target variable and created tiers for our model to predict. However, wRC+ is continuous, and partitioning it removes that characteristic. When our model tries to classify our data, it could have some trouble with data that falls near the artificial boundaries that we created, thus hurting our model’s accuracy.

Clustering results were weak on the surface, but revealed player archetypes after diving into each cluster.

We have features with high correlation (multicollinearity). We removed some of these features; however, our models without feature selection outperformed models with RFE, suggesting our ensemble models were able to deal with the multicollinearity.

We detected some outliers; however, we wanted to confirm whether they are real performances or erroneous data. We cross-referenced the outliers using domain knowledge to verify that the outliers were indeed real data, leading us to keep the outliers.